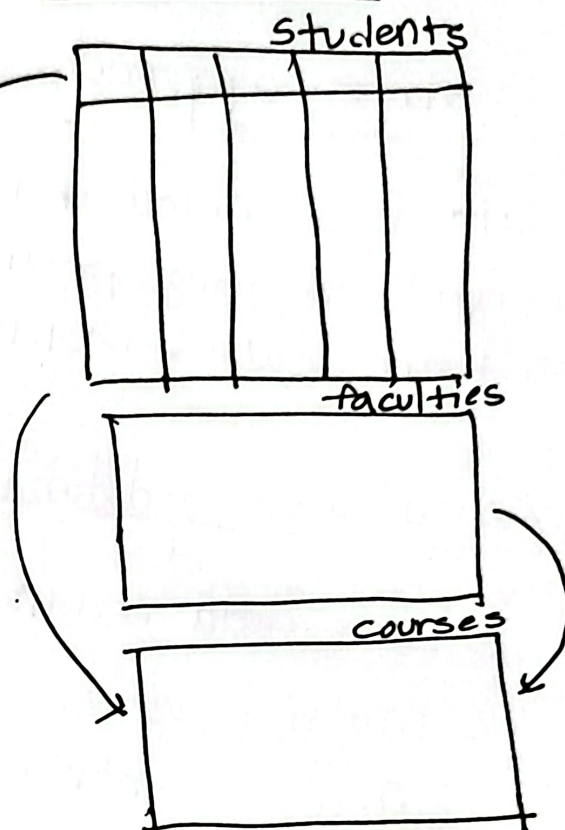


Data

└─ Unstructured ✗ (not covered)
└─ Structured ✓

col rows
↓ ↓
properties instance
 of object

Relational database



SQL → structured Query Language

DBMS → MySQL, oracle
Database Management systems

Data → Known facts that can be recorded and have an implicit meaning

Database

→ represents some aspect of real world or a mineworld

→ logically coherent collection of data with some inherent meaning

DBMS

→ software package to facilitate creation and maintenance of a computerized database

Database system

→ DBMS software with data and UI

Managing data — { File based approach Database approach

Database approach

Data is collected and manipulated using specific software called DBMs.

File based approach

Utilizes a collection of application programs which performs services to end users.

Advantages of database approach ↓ redundancy

- | | |
|-------------------------|--------------------------|
| 1. Less data redundancy | 6. complex Relationships |
| 2. Multiple views | 7. Integrity constraints |
| 3. Access control | 8. Optimized queries |
| 4. concurrency control | |
| 5. Backup and recovery | |

Disadvantages of database approach

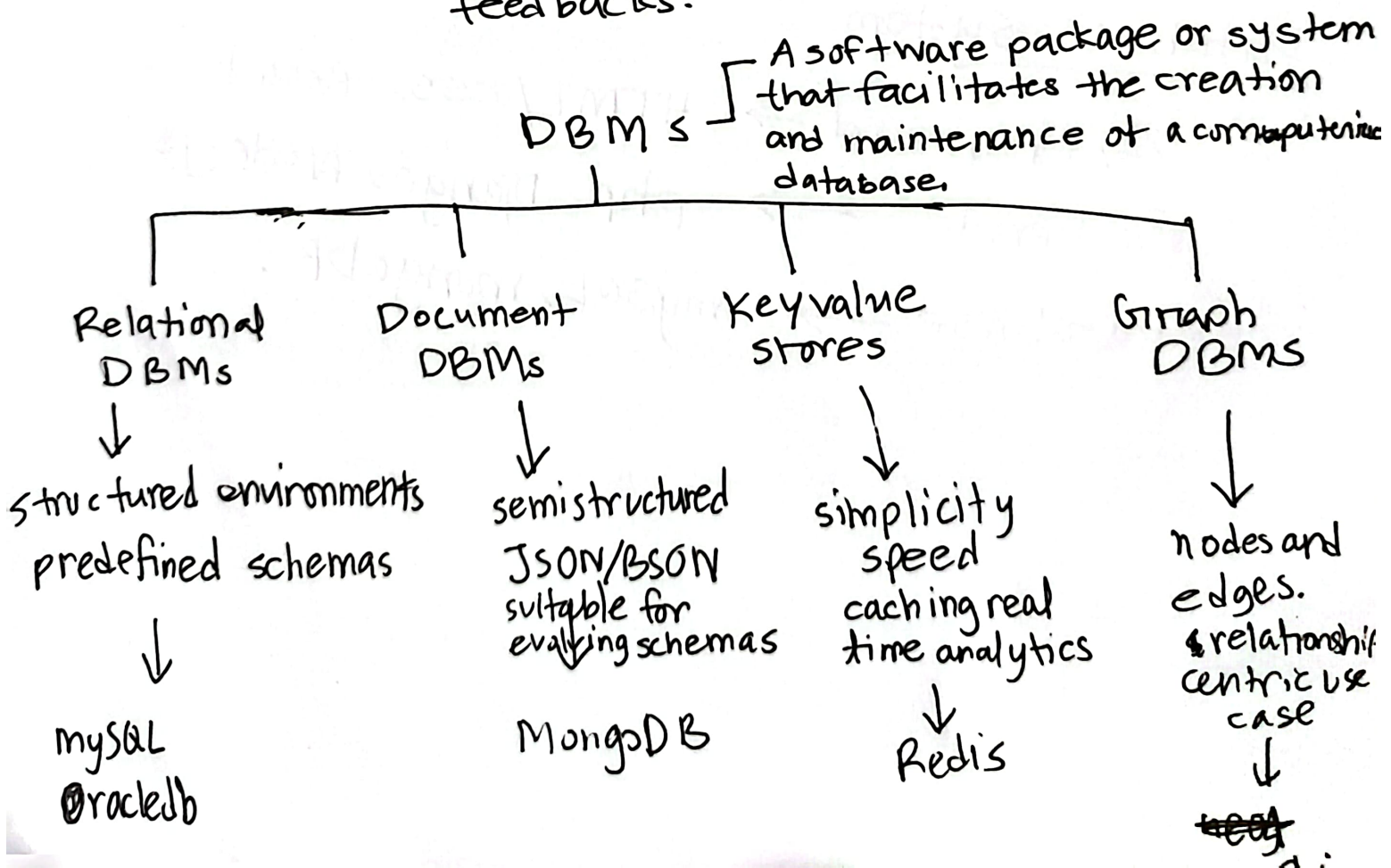
1. High initial cost - software + setup cost + developer cost
2. Extra hardware maybe needed
3. Overhead → managing security, recovery, concurrency etc.
4. Maintenance cost → need to hire technical support for system maintenance

Database designers → designs database by meeting requirements from end users & stakeholders. define structure and constraints of the database.

Software engineers → Implements the database using a DBMS and develop apps that interact with db using programming languages.

Database admins → They monitor and control the use and performance of database. perform backups, system updates and troubleshooting.

End users → interacts with the database, update data through web apps & provide feedbacks.



DBMS Languages

DDL (Data Definition Language)

Defines/alters the database structure/schema. E.g. create a table, add/remove columns, change column names, data types or format etc.

SQL → create Table, Alter Table

DML (Data manipulation Language)

modifies/retrieves data by inserting new records/rows; updating existing values; deleting existing records/rows and retrieving stored data.

SQL → Insert Into, Delete from, Update, Select

Database System

UI/Front end → HTML/CSS, React

Backend → php, Django, Node.js

Database → MySQL, MongoDB.

Database Schema

↳ description of a database.

↳ includes —

- description of the database structure
- data types
- constraints

* to change/create the schema of a database DDL queries are required.

Database State

Actual data stored in a database at a particular moment in time. (snapshot/occurrence)

* to change/view the state of a database DML queries are required.

Three Schema Architecture

Internal schema → physical storage structures & access paths

Conceptual schema → structure & constraints for the whole database for a community or others

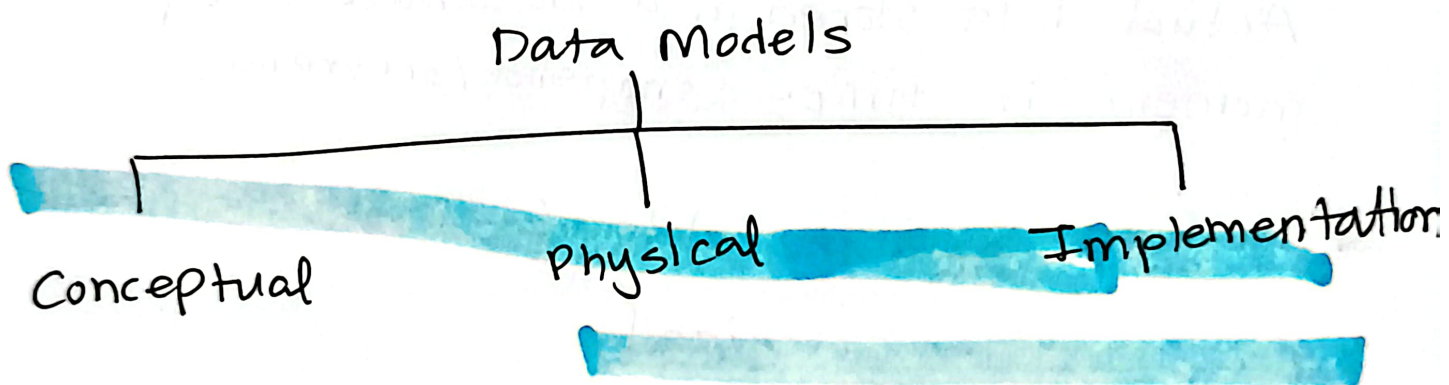
External schema → At the external level, to describe various user views

Logical Data Independence

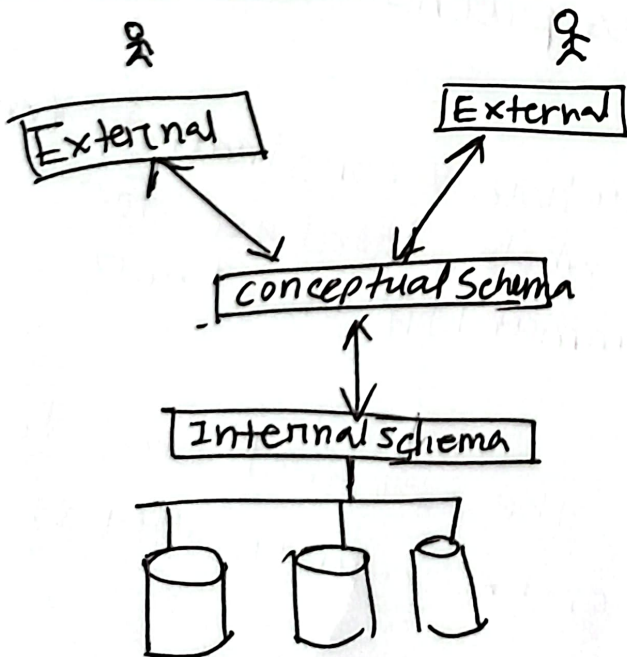
Capacity to change the conceptual schema without having to change the external schema.

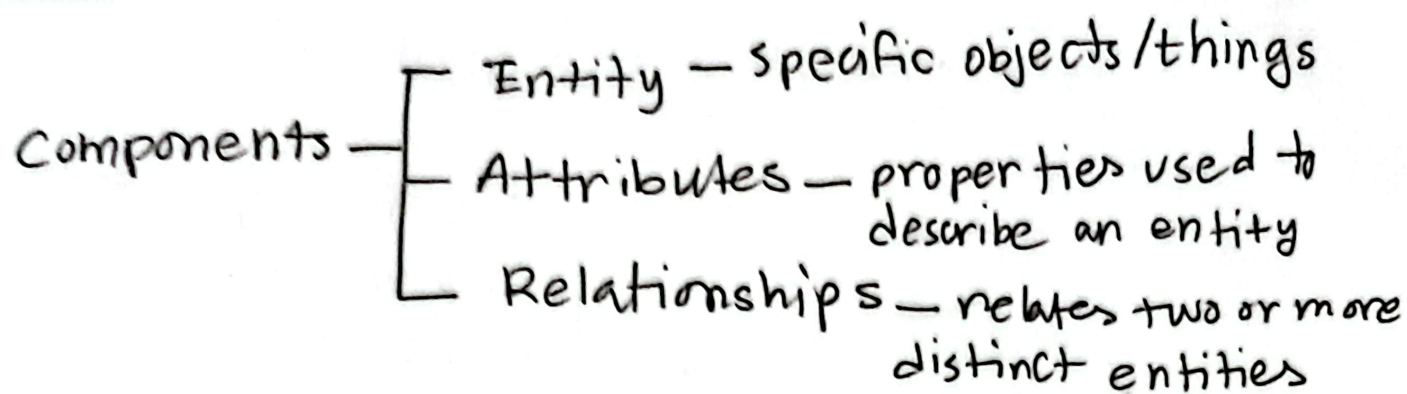
Physical data independence

Capacity to change the internal schema without changing the conceptual schema.

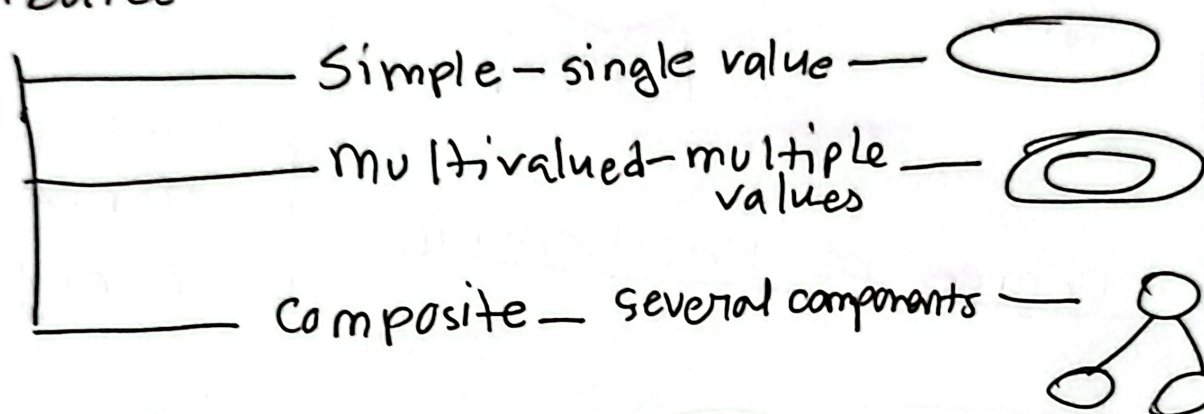


3 schema architecture

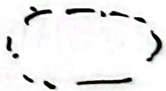


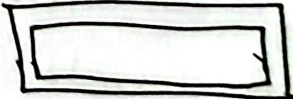


Attributes



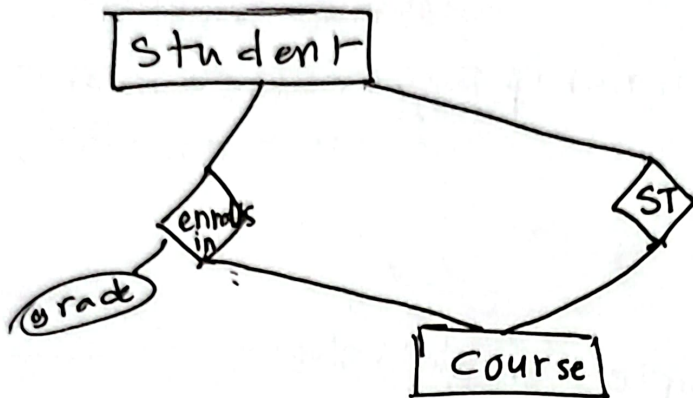
* Key Attribute — unique value —  underlining must

* Derived attribute — calculated / derived from other stored data — 

Weak entity — (no unique key) — 

Partial key — part of a unique key — 

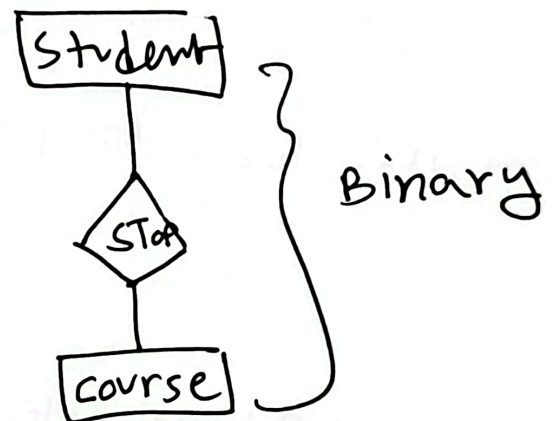
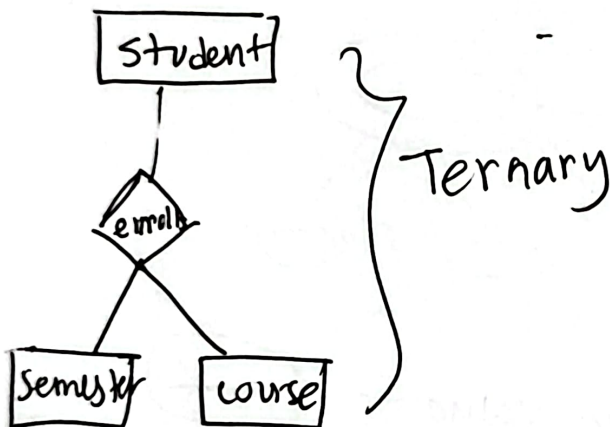
Relationships — relates two or more specific entities with a specific meaning.



Relationship types may or may not have attributes.

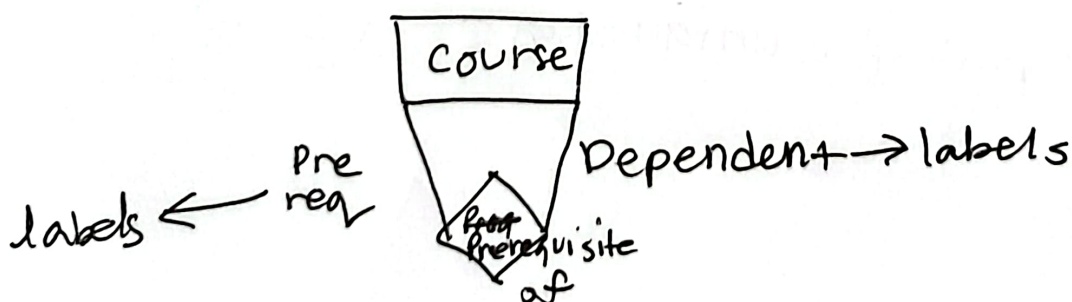
Degree of Relationship —

- Binary (2 participants)
- Ternary (3 participants)
- n-ary (n participants)



Recursive Relationship

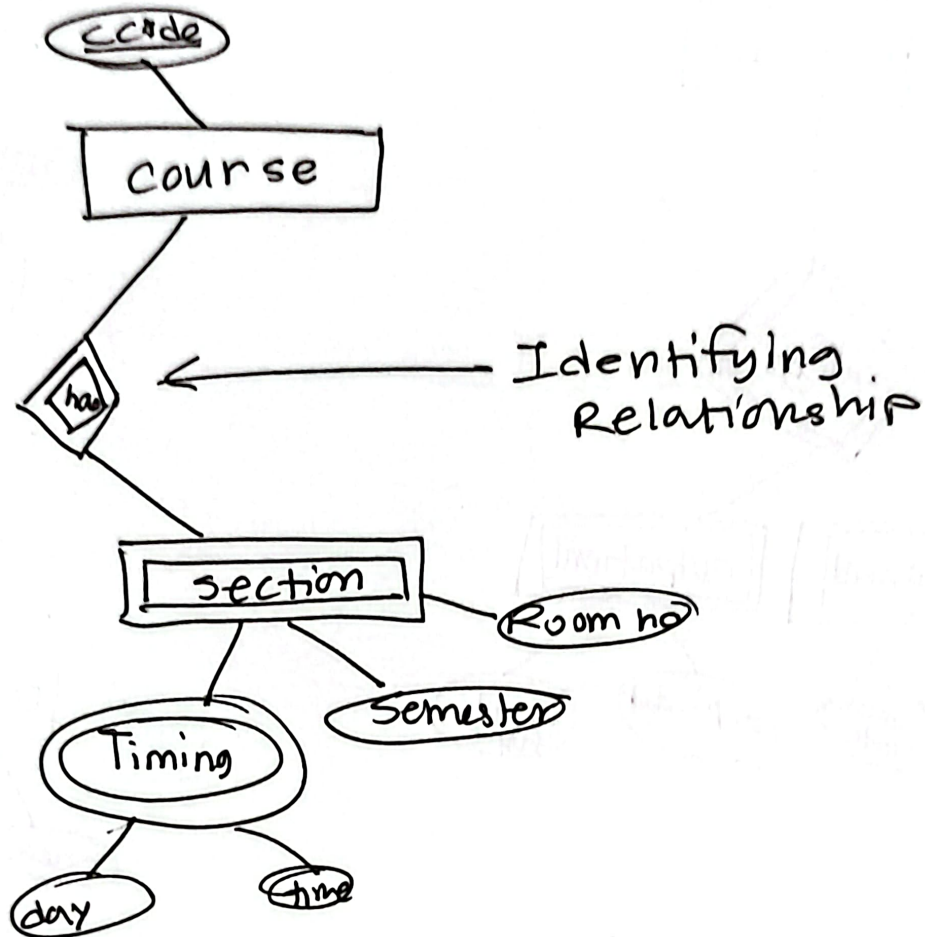
Two entities of the same entity type participate in a relationship.



Identifying Relationship

- 15

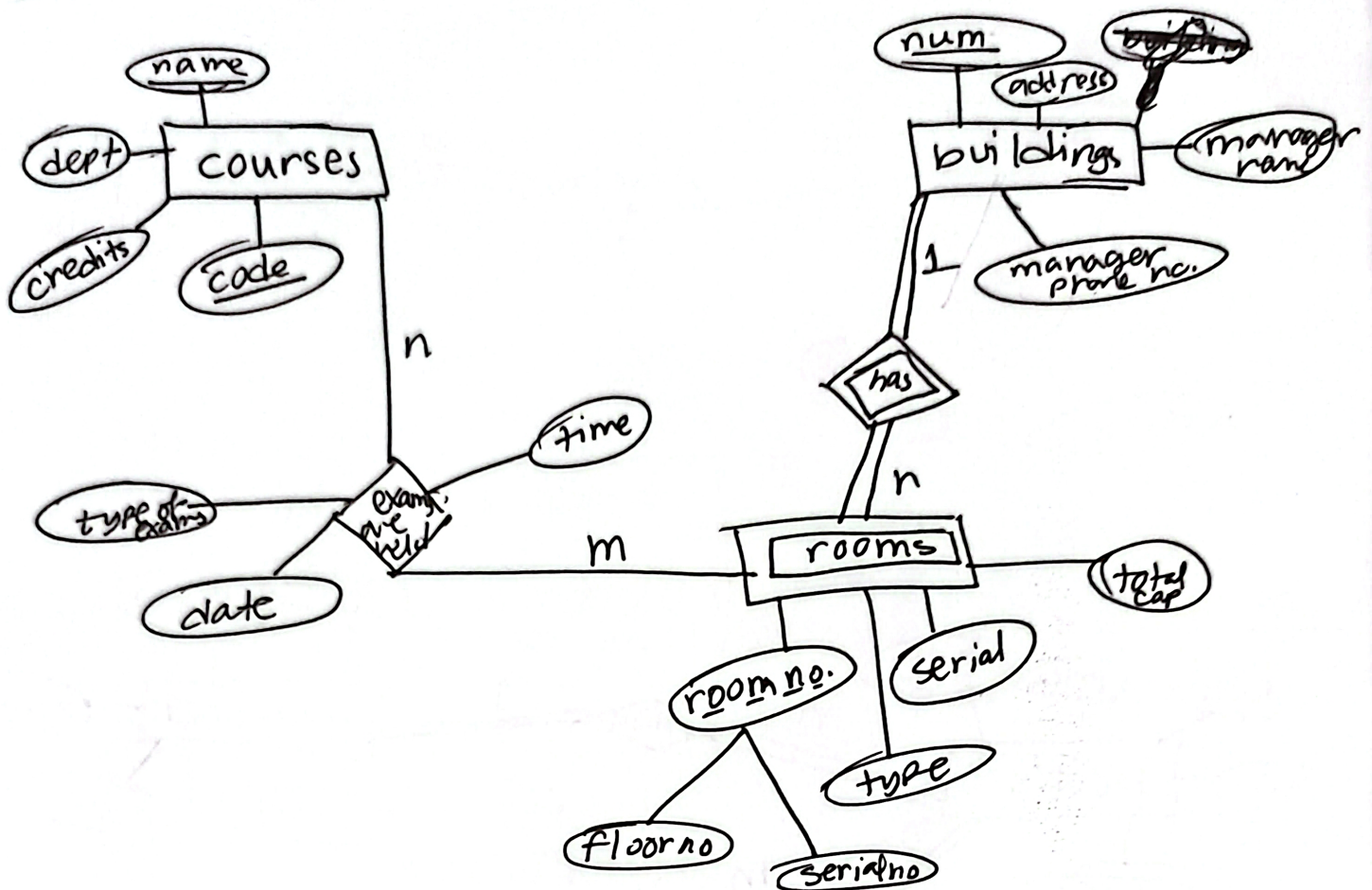
A weak entity must participate in an Identifying Relationship with an owner or identifying entity type.



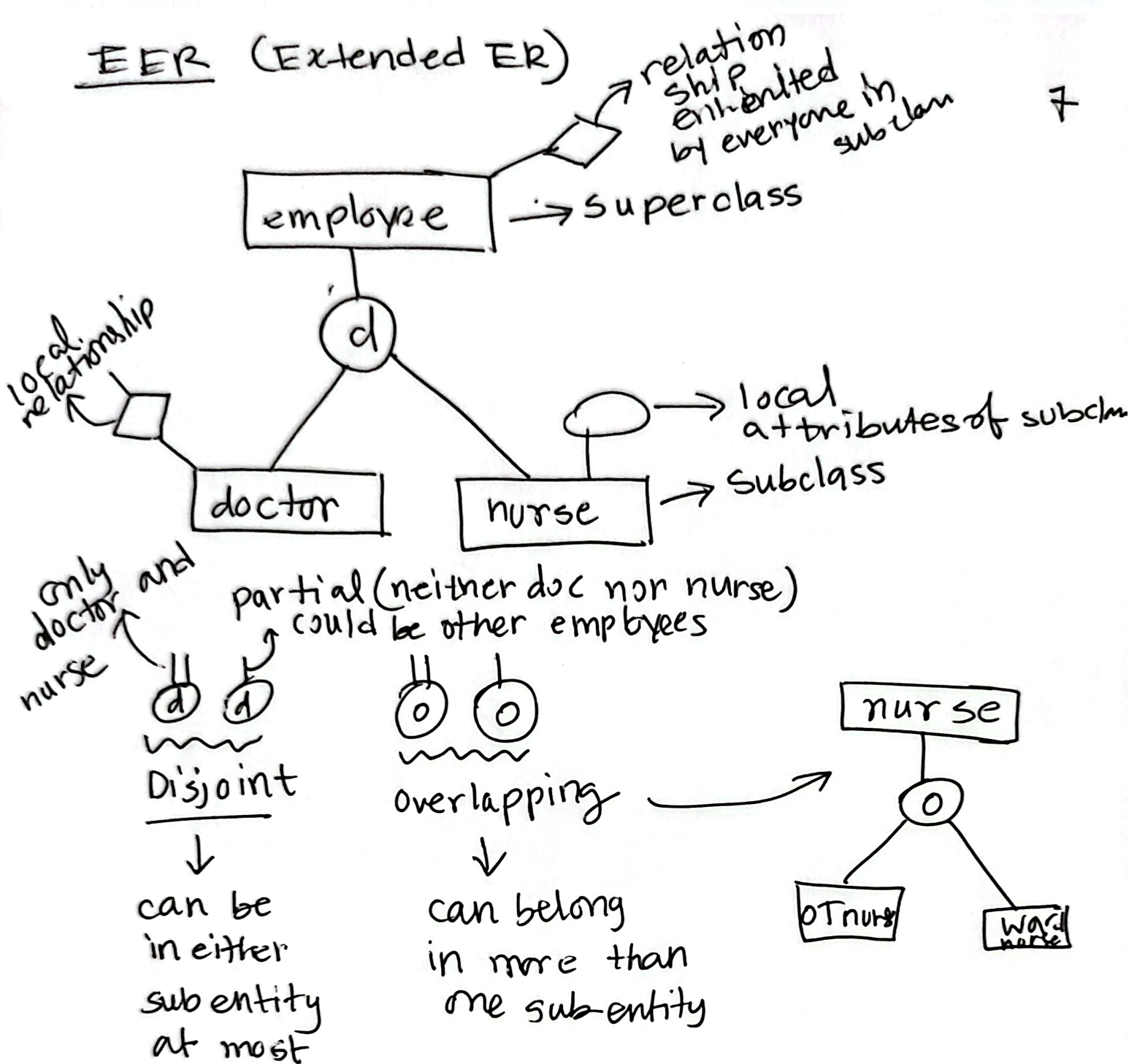
Constraints

* Cardinality Ratio $\rightarrow 1:1, 1:n, m:n$

* Participation constraint $\rightarrow$ Optional $\rightarrow$ —
 $\rightarrow$ mandatory $\rightarrow$ ==



EER (Extended ER)



Generalization → subentity (210) superclass
 bottom up

Specialization → superclass (210) subentity
 top up

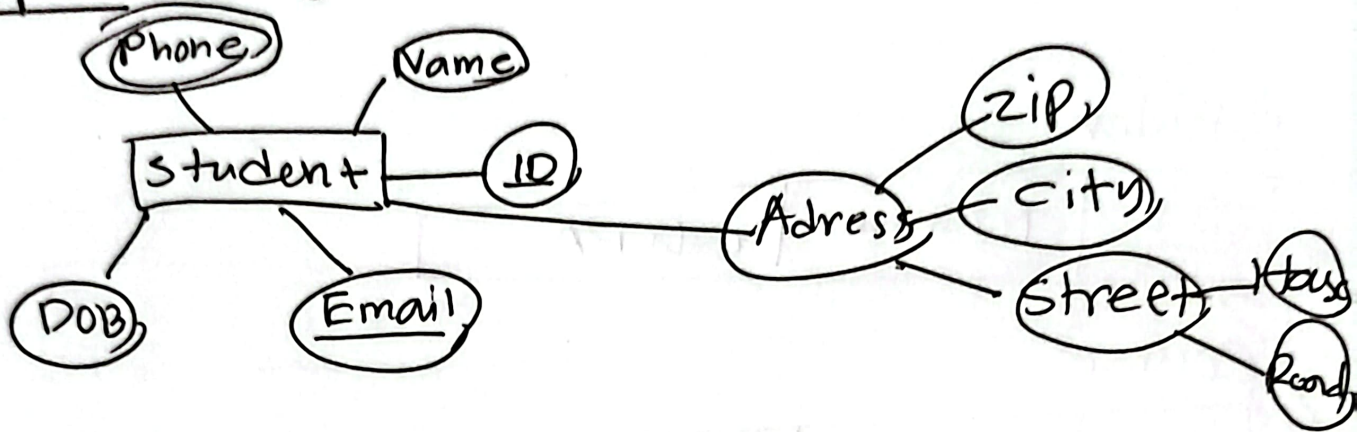
Schema Mapping

8

ER → Step 1-7

EER → Step 7-8

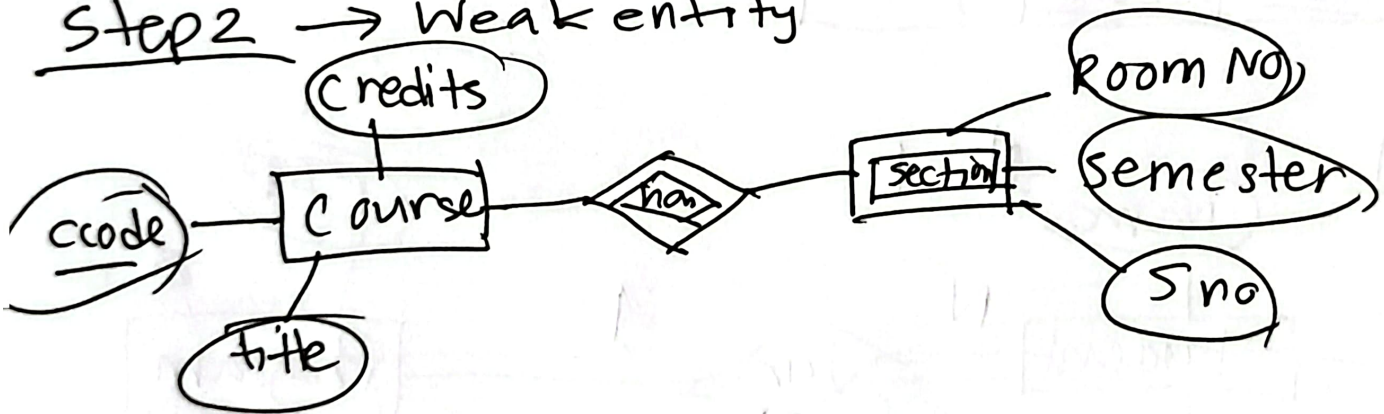
Step 1 → Regular Entity Type



Student

<u>\$ID</u>	Name	Email	DOB	City	House	Zip	Road
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Step 2 → Weak entity



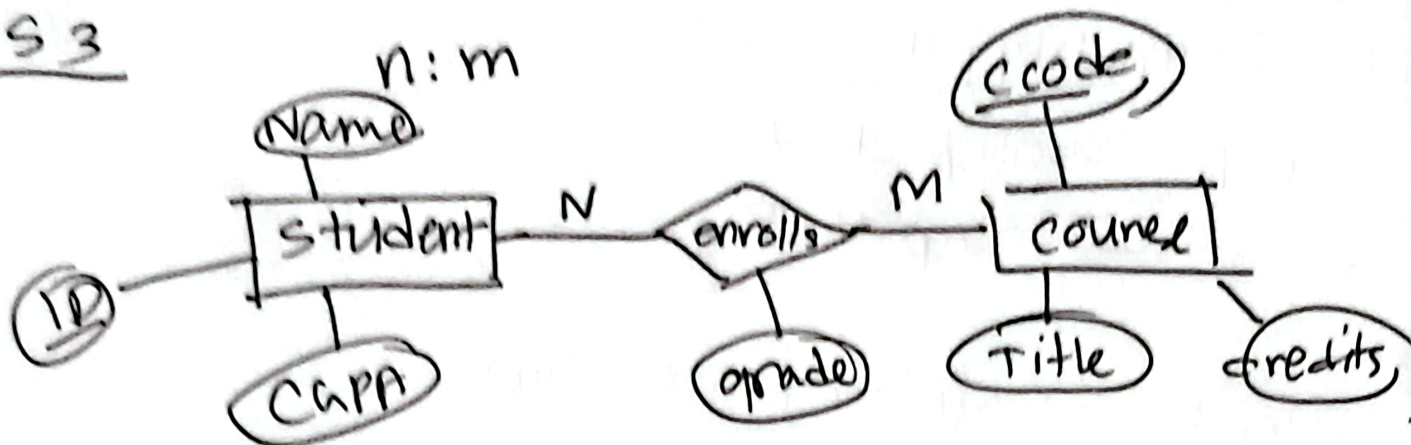
Course

<u>ccode</u>	title	Credits
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section

Room No.	<u>Sno</u>	Semester	ccode
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S3



Student

<u>ID</u>	Name	CGPA
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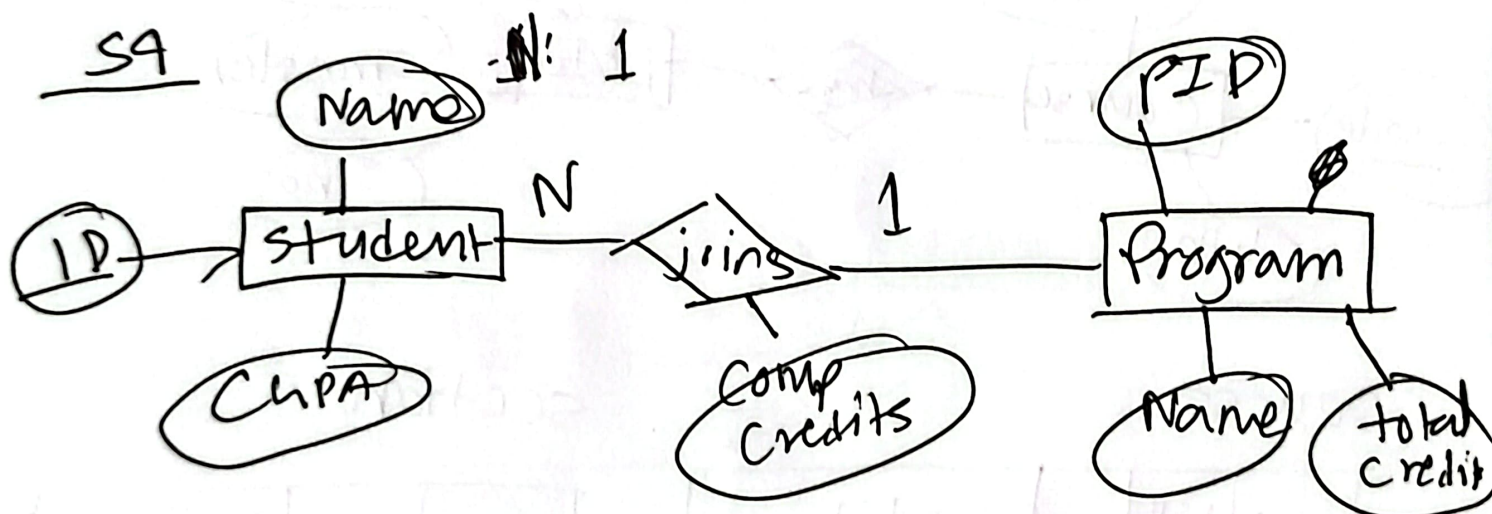
Course

Title	credits	<u>ccode</u>
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enrolls

<u>ID</u>	<u>ccode</u>	grade
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S4

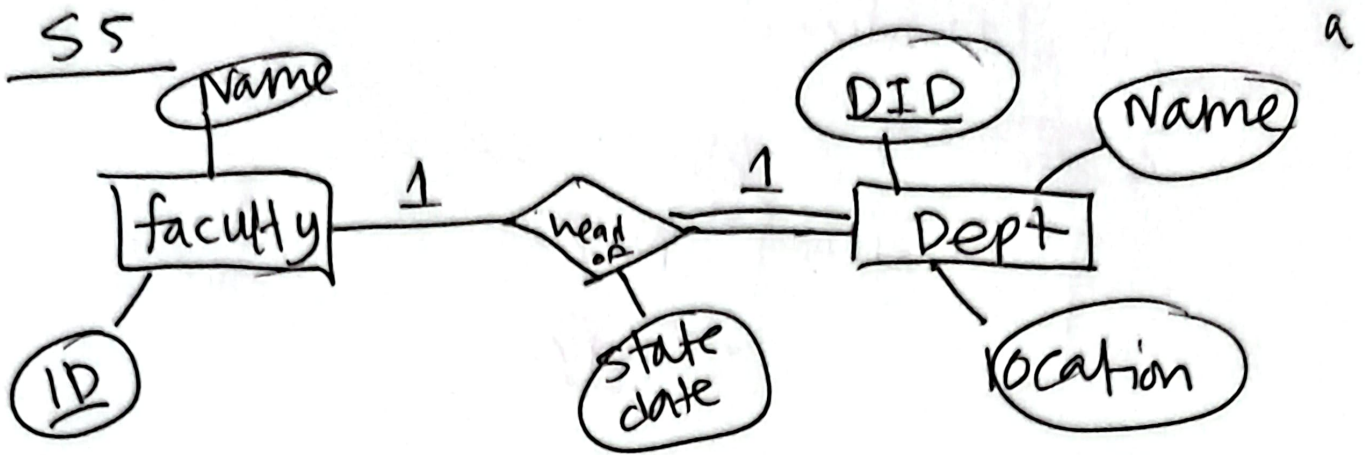


Student

<u>ID</u>	CGPA	Name	Comp. Credit	PID
-----------	------	------	--------------	-----

Program

<u>PID</u>	Name	total credit
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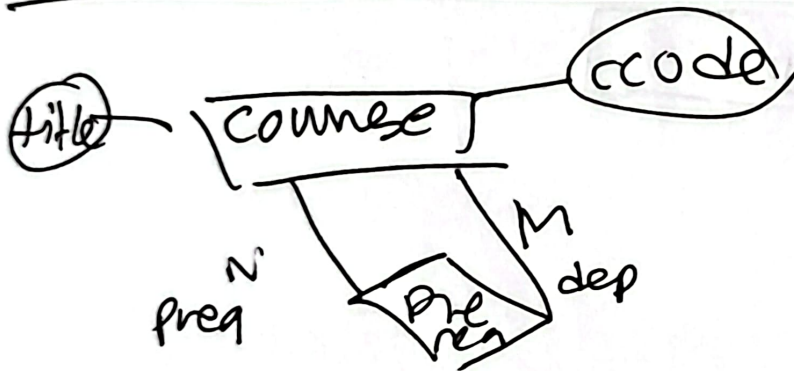
Faculty

<u>ID</u>	Name
-----------	------

Dept

<u>DID</u>	Name	loc	start date	FID
------------	------	-----	------------	-----

Recursive R/M same as n:m



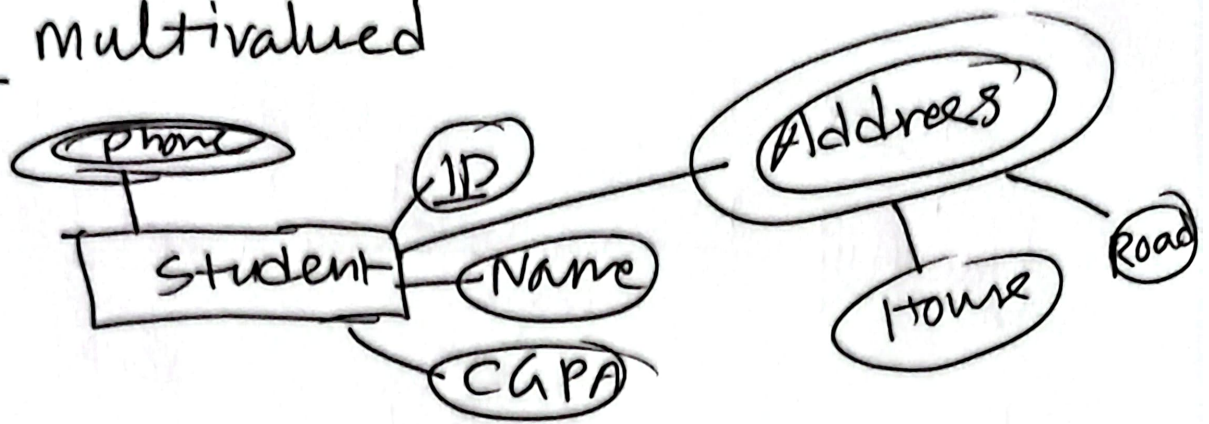
Pre req

<u>N code</u>	<u>M code</u>
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Course

<u>ccode</u>	title
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S6 multivalued



Student

<u>ID</u>	Name	CGPA
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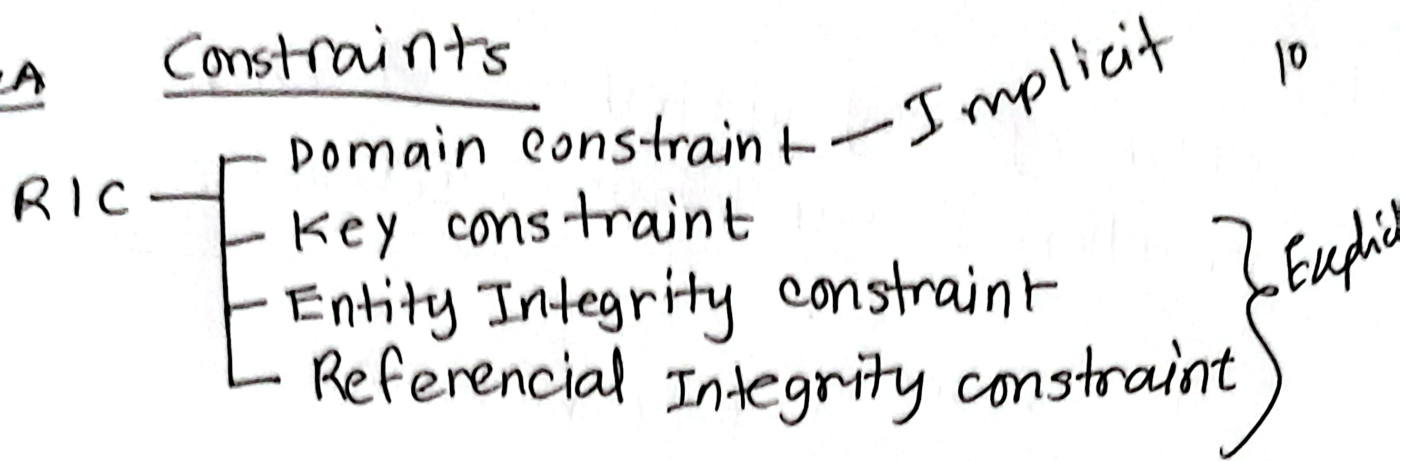
Phone

<u>Phone</u>	<u>ID</u>
--------------	-----------

Address

<u>house</u>	road	<u>ID</u>
--------------	------	-----------

DBA Constraints 10



Domain constraint (Implicit)

Domain $\rightarrow$ set of all possible values for each attribute

যদি domain এর বাইরে হয় তাহলে domain constraint violation.

e.g. Age takes 0-100 values so if somewhere we insert age as -3
Then dcv

Key constraint (Explicit)

যদি key attribute এর value duplicate হয়, [Primary key unique হবে]

10	name	} violation
1	Tanzim	
1	Anik	

যদি প্রত্যেক Primary key unique হয় ওহলে
একটি diff হয়.
no violation

ETC

primary key attributes of each relation R can't have null values

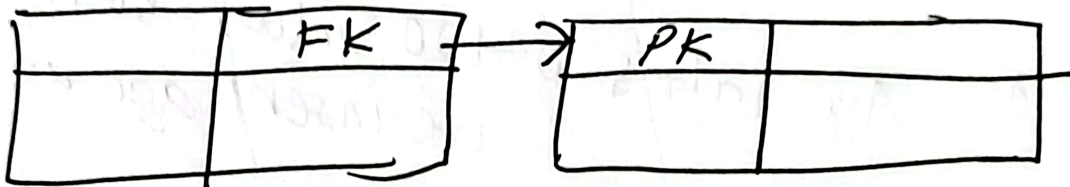
ID	name
1	Tanzim
null	Anik

 } ETC violation

note { যদি null value আসত না তবে but তখনও
আসত বসত then it won't be a violation
as it's not primary key

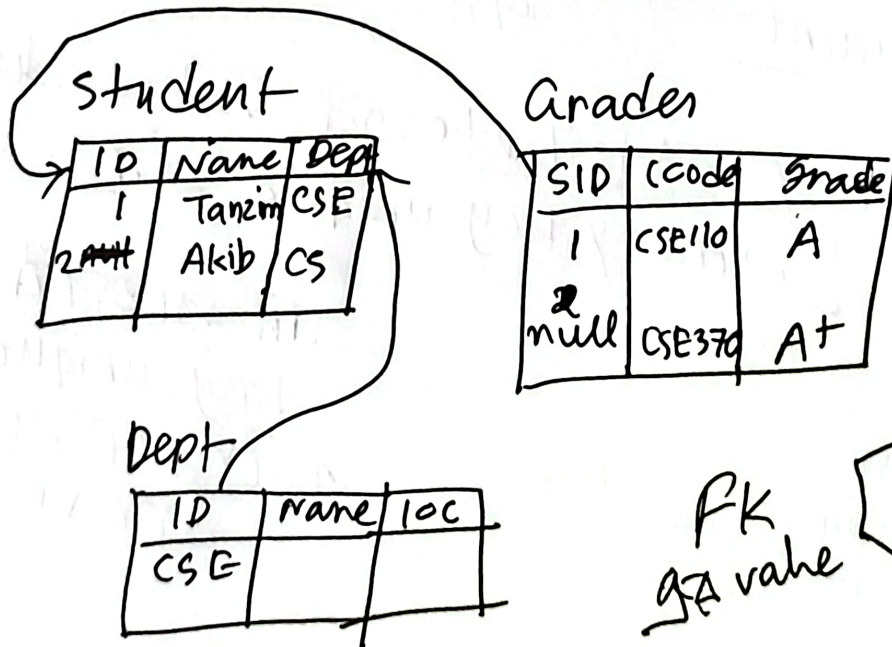
RIC

Arrow দিবে FK $\rightarrow$ PK



Referencing
table

Referenced
table



FK
ga value

existing
value
in student
table ID
null
only if the
foreign key

~~Insert~~

OP	DC Insert	KC Delete	FKC	R
Insert	data type must be specified.	declare primary key & unique keyword.	declare primary key	de fb pro rel
Delete	X	X	X	Res casc set n
Update	Same as insert	Same as insert	Same as insert	Sam as delet

Restrict → Doesn't allow deletion from referenced table

Cascade → Deletes all rows from referencing the deleted primary key

Set Null → sets fk to null